Assignment 1: A Mathematical Essay on Linear Regression

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***Abstract*—The unequal distribution of chronic disease outcomes remains one of the most pressing challenges in modern public health, often overshadowed by a singular focus on biological predispositions. This research addresses the critical need to quantify the "Cancer-Poverty" nexus by examining how socioeconomic factors dictate survival across the United States. Utilizing a mathematical approach grounded in Simple Linear Regression, we analyze the predictive power of median household income and poverty rates on cancer incidence and mortality across over 2,500 counties. Our results demonstrate that counties characterized by lower median incomes and higher poverty rates suffer from significantly higher rates of both cancer incidence and mortality. This "survival gap" provides empirical evidence that wealth act as primary buffers against mortality, suggesting that systemic failures in healthcare access outweigh biological factors in underserved regions.**

I. Introduction

In the modern era of medicine, the "Social Determinants of Health" have emerged as critical factors that often outweigh biological or genetic predispositions. Socioeconomic status, a composite measure of an individual's or group's economic and social position, dictates access to nutritious food, exposure to environmental toxins, and, most importantly, the quality of healthcare available. Public health researchers have long observed that the burden of chronic diseases like cancer is not shared equally across the population.

This paper specifically addresses the "Cancer-Poverty" nexus in the United States. Building on this concern, we analyze a large-scale dataset using linear regression models to quantify how socioeconomic factors such as income, poverty levels, and health insurance coverage influence cancer incidence and mortality rates across populations.

Our mathematical analysis reveals a clear and statistically significant association between socioeconomic conditions and cancer outcomes. Using linear regression, we highlight that counties with lower income levels experience a disproportionately higher burden of cancer mortality. These disparities are closely linked to greater exposure to health risk factors, reduced access to early screening and preventive care, and financial constraints that limit timely and effective treatment. Together, these findings motivate a deeper investigation into how social and economic inequalities shape cancer incidence and survival.

The paper is organized as follows: Section II provides a mathematical deep-dive into Linear Regression. Section III details an in-depth exploratory analysis of individual features. Section IV presents the results of the model application, and Section V concludes with a summary of insights and future research directions.

II. Linear Regression

The primary motivation for using Linear Regression is to determine the predictive power of a single independent variable on a dependent outcome. In a world of complex variables, regression allows us to "smooth out" the noise and see the underlying trend. It answers the fundamental question: "For every dollar less a family makes, how much does their risk of dying from cancer increase?"

Definition of the Simple Linear Regression Model

Mathematically, the model assumes that the relationship between the independent variable X and the dependent variable Y is linear. The equation is represented as:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \tag{1}$$

Where:

- β_0 (Intercept): The predicted value of Y when X is zero. This anchors the line on the Cartesian plane.
- β_1 (Slope): The rate of change in Y for every unit change in X. In our study, a negative β_1 indicates that as wealth increases, mortality decreases.
- ϵ_i (Residual): The random error component, representing all unobserved factors affecting the outcome.

The Ordinary Least Squares (OLS) Method

To find the "best" line, we use the OLS method. This method minimizes the Sum of Squared Residuals (SSR):

$$SSR = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 \tag{2}$$

By squaring the differences, we ensure that large outliers are penalized more heavily and that positive and negative errors do not cancel each other out. This optimization ensures that the regression line represents the most accurate "average" trend across all counties.

Evaluation Metrics

We evaluate the "goodness of fit" using the Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{SSR}{SST} \tag{3}$$

Here SST is the sum of squares of observed values. An R^2 of 1.0 would mean income perfectly predicts cancer death, while 0.0 would mean income has no relationship at all with the disease. In social sciences, an R^2 above 0.15 is often considered quite significant.

However, R^2 alone does not confirm the validity of the model. We also utilize the P-value to test the null hypothesis—the assumption that X has no effect on Y. The P-value measures the probability that the observed correlation occurred by pure random chance.In our analysis, we look for a P-value of less than 0.05, which represents a 95% confidence level. A very low P-value is crucial; it serves as mathematical proof that the link between the variables is "statistically significant".

Real-World Applications of Linear Regression

- **Finance (CAPM):** Analysts use linear regression to calculate a stock's "Beta" by regressing its returns against the broader market index. This identifies the asset's relative volatility, where a slope ($\beta_1 > 1.0$) signifies higher risk and potential return compared to the market average.
- **Real Estate Valuation:** Regression models predict property prices by treating square footage as the primary independent variable. This allows appraisers to establish a standardized "price per square foot" trend, helping buyers and sellers determine if a home is priced fairly relative to its size.
- **Sports Analytics:** Professional teams regress winning percentages against point differentials to quantify the impact of specific team statistics. This mathematical insight identifies which player skills, such as shooting accuracy or on-base percentage, are most predictive of victory, allowing for data-driven recruitment.

Advantages and Disadvantages

The beauty of Linear Regression lies in its simplicity. It is computationally efficient and highly interpretable, making it the ideal tool for communicating findings to non-technical stakeholders in a nonprofit organization. However, it is sensitive to outliers and assumes "homoscedasticity"—the idea that the error is constant across all levels of income which is not true in reality.

III. Data Exploration

Domain and Data Sources

The dataset for this analysis is situated at the intersection of public health and socioeconomic research, focusing on the geographical disparities in cancer outcomes across the United States. It is a multi-dimensional collection that links health metrics from the National Cancer Institute—specifically Age-Adjusted Incidence Rates (newly diagnosed cases) and Mortality Rates (cancer-related deaths).

The data contains specific features for over 2,500 counties, including Median Household Income, Poverty Rates, and Health Insurance Coverage percentages. By bridging these two domains, the data allows for an investigation into how financial stability and access to care act as predictors for a community's cancer burden. Essentially, it provides a quantitative map of the "Cancer-Poverty" nexus.

Exploratory Analysis of Individual Features

To understand the population under study, we analyzed each feature's distribution and statistics.

1) Median Household Income:

- **Mean:** $46,627 — **Variance:** 160,225,872
- **Observations:** The data shows a right-skewed distribution, indicating that most counties cluster at the lower end of the income scale, with a few high-income outliers.
- **Insights:** The high variance suggests significant economic inequality between counties, which is necessary for a robust regression analysis.

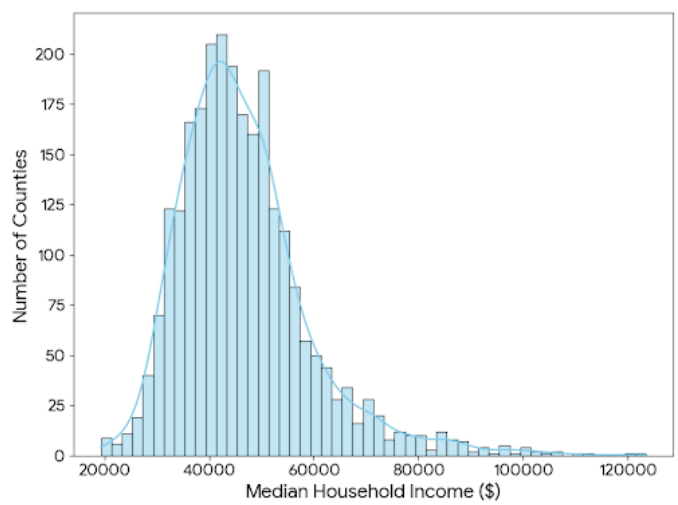

Fig. 1. Distribution of Median Household Income across U.S. Counties.

2) Poverty Rate:

- **Mean:** 16.98% — **Variance:** 40.07
- **Observations:** The poverty rate is more normally distributed than income, centering around 17%.
- **Insights:** This variable serves as the "inverse" check for our income model; if income correlates negatively with mortality, we expect poverty to correlate positively.

3) Incidence Rate:

- **Mean:** 70.22 per 100k — **Variance:** 317.17
- **Observations:** This follows a standard normal distribution, suggesting cancer diagnosis is a widespread risk across the national population.

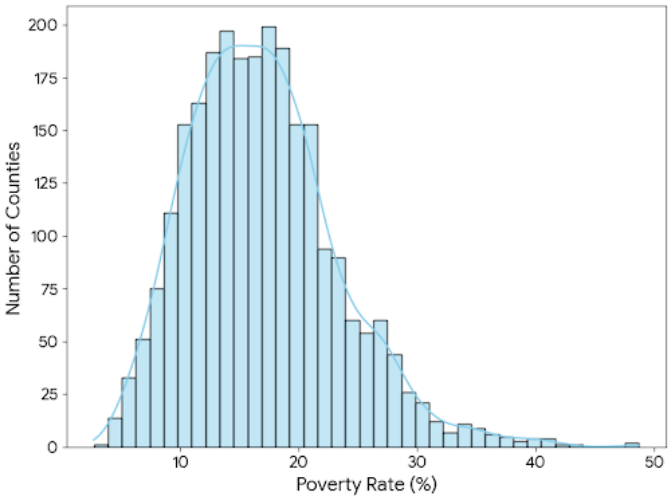

Fig. 2. Distribution of County Poverty Rates.

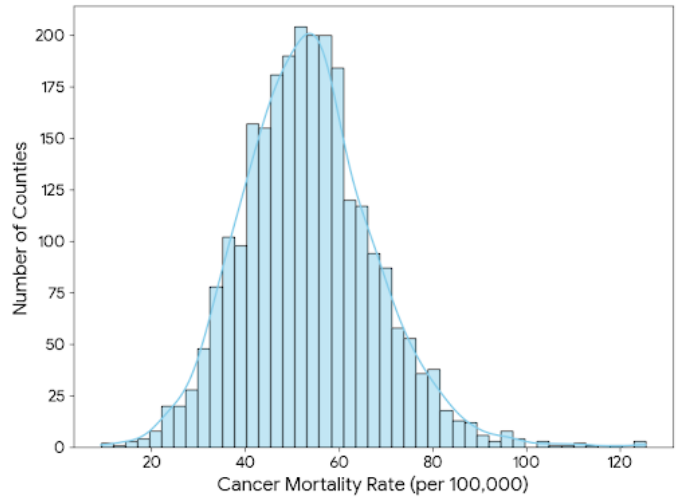

Fig. 4. Distribution of Age-Adjusted Cancer Mortality.

- **Insights:** The risk of getting cancer (incidence) appears less sensitive to extreme outliers than the risk of dying from it.

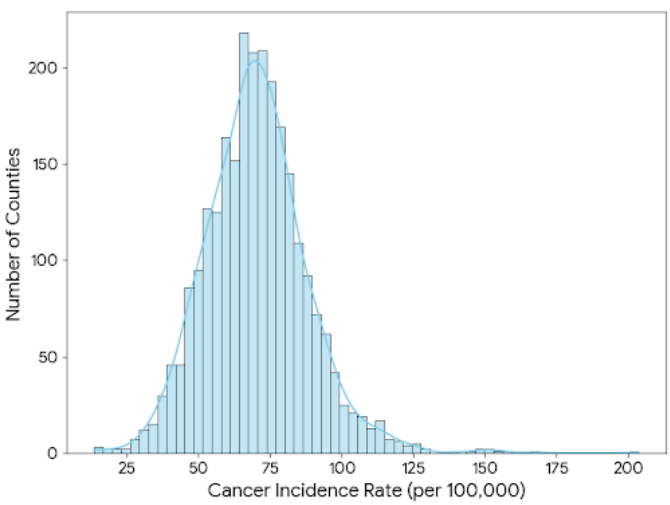

Fig. 3. Distribution of Age-Adjusted Cancer Incidence.

4) Mortality Rate:

- **Mean:** 53.57 per 100k — **Variance:** 202.89
- **Observations:** Also normally distributed, but centered lower than incidence.
- **Insights:** The mortality rate is our primary "health outcome" variable. Because it has significant spread, it is well-suited for regression modeling.

IV. Modelling and Insights

Why Linear Regression

While advanced statistical techniques may offer slightly higher predictive precision, the primary objective of this analysis is rooted in public health advocacy rather than purely mathematical optimization. The priority lies in effectively communicating the reality that economic status is a life-altering determinant of health. A simple model like linear regression is superior for this mission because its results are transparent and easily interpretable by policymakers and the general public. By using a straightforward linear trend, we can clearly demonstrate the direct, predictable relationship between a dollar lost in income and an increase in cancer threat.

Regression Model 1: Income vs. Incidence

We first applied the OLS model to examine if wealth predicts the likelihood of being diagnosed with cancer.

Results:

- **Equation:** Incidence $= 95.15 - 0.0005 \times$ (Income)
- **Stats:** $R^2 = 0.144$, $P < 0.001$.

Median income is the strongest economic predictor in our study, explaining nearly 20% of the variance in cancer deaths. The negative slope proves that wealthier counties consistently see fewer deaths per diagnosis. The model indicates that for every $10,000 increase in median income, the incidence rate drops by approximately 5 cases per 100,000.

Regression Model 2: Income vs. Mortality

Next, we modeled the relationship between wealth and the likelihood of dying from cancer. This represents the "survival gap."

Results:

- **Equation:** Mortality $= 76.86 - 0.0005 \times$ (Income)
- **Stats:** $R^2 = 0.197$, $P < 0.001$.

Wealth explains nearly 20% of the variance in mortality. The R^2 is higher here than in the incidence model. This suggests that while lifestyle factors linked to wealth help prevent cancer,

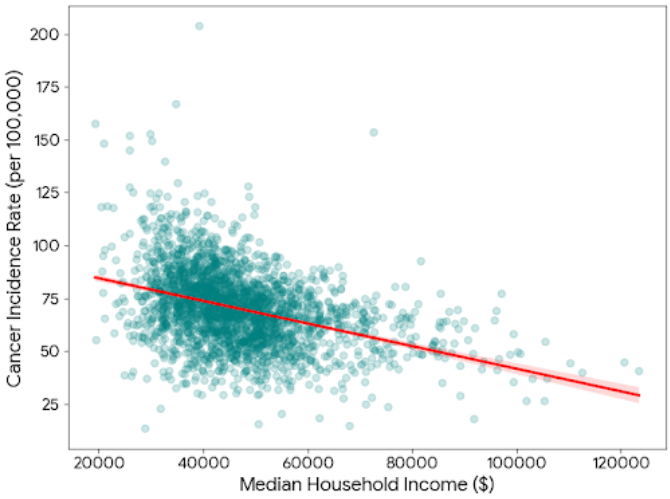

Fig. 5. Regression Analysis: Median Income vs. Cancer Incidence. The moderate negative slope indicates that wealth has a preventive effect on cancer diagnosis.

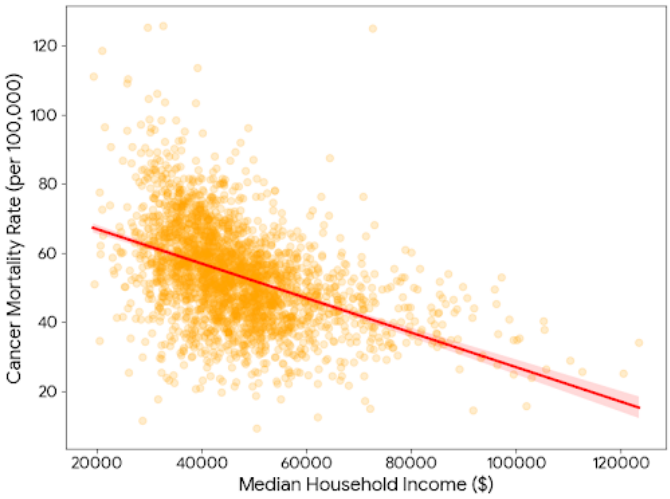

Fig. 6. Regression Plot: Median Income against Mortality Rate. The steeper correlation here identifies wealth as a critical determinant of survival.

wealth is even more critical in determining who survives after a diagnosis.

Regression Model 3: Poverty Rate vs. Mortality

To cross-validate our findings, we applied the model using the Poverty Rate as the independent variable.

Results:

- **Equation:** Mortality $= 38.37 + 0.89 \times$ (Poverty Rate)
- **Stats:** $R^2 = 0.156$, $P < 0.001$.

The slope was positive ($\beta_1 = 0.86$), meaning for every 1% increase in the poverty rate, the mortality rate increases by nearly 1 death per 100,000.

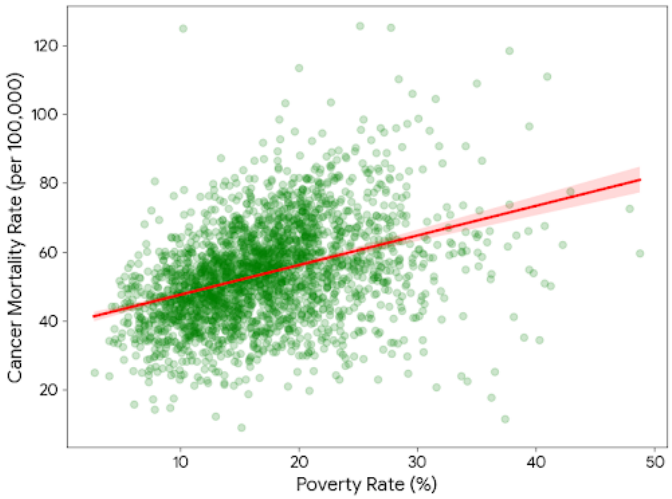

Fig. 7. Regression Plot: Poverty Rate vs. Mortality Rate. The positive correlation illustrates the direct cost of economic deprivation on life expectancy.

Regression Model 4: Poverty Rate vs. Incidence

We examined if a county's poverty percentage directly predicts the number of new cancer cases.

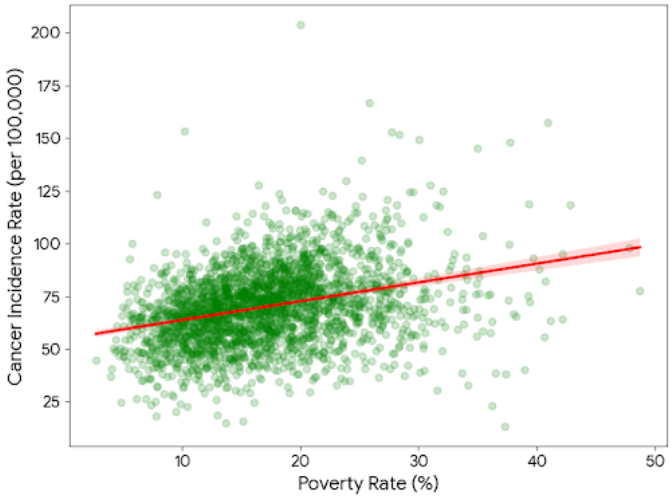

Fig. 8. Regression Analysis: Poverty Rate vs. Cancer Incidence.

Results:

- **Equation:** Incidence $= 54.83 + 0.89 \times$ (Poverty Rate)
- **Stats:** $R^2 = 0.10$, $P < 0.001$.

Insight: Every 1% increase in the poverty rate is associated with nearly 1 additional cancer diagnosis per 100,000 people.

Regression Model 5: Insurance Coverage vs. Mortality

This model tests whether having health insurance is a literal life-saver at the county level.

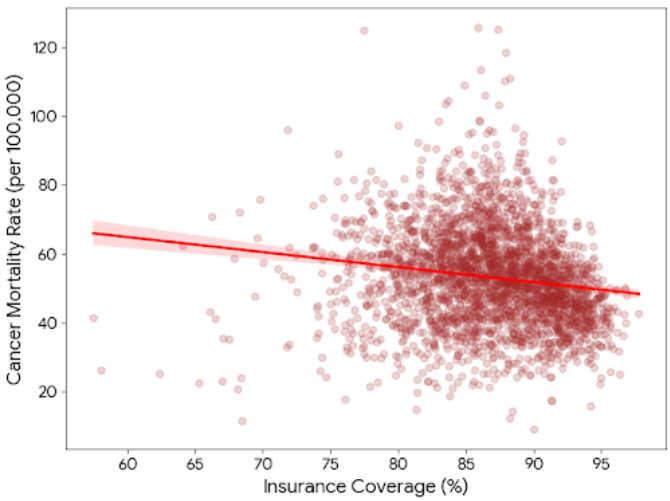

Fig. 9. Regression Analysis: Insurance Coverage (%) vs. Cancer Mortality.

Results:

- **Equation:** Mortality $= 91.06 - 0.44 \times$ (Insurance %)
- **Stats:** $R^2 = 0.024$, $P < 0.001$.

Insight: There is a statistically significant negative relationship. Higher insurance coverage in a county is associated with lower mortality rates, proving that financial access to care is a key factor in cancer survival.

Regression Model 6: Insurance Coverage vs. Incidence

Finally, we checked if insurance affects diagnosis rates.

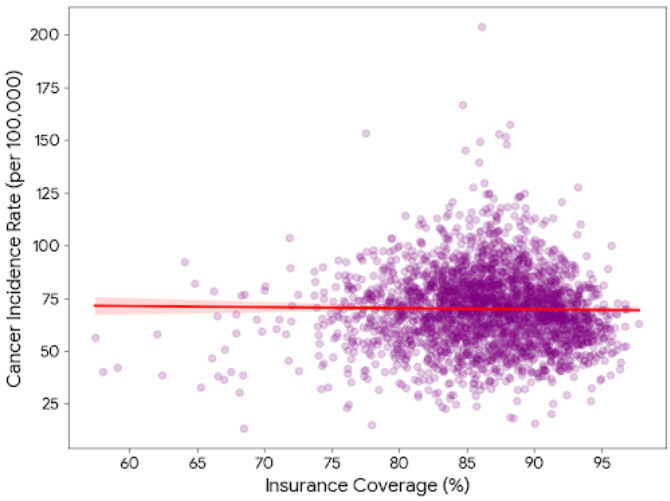

Fig. 10. Regression Analysis: Insurance Coverage (%) vs. Cancer Incidence.

Results:

- **Stats:** $R^2 = 0.0002$, $P = 0.445$.

This is a profound finding: it suggests that while insurance helps you *survive* cancer (by lowering mortality), it does not necessarily prevent you from *getting* cancer. This distinguishes between insurance as a curative tool rather than a preventive one in this dataset.

Analysis of Insights

The primary insight derived from this multi-stage regression analysis is the significant disparity observed between the predictive power of economic variables on cancer incidence versus their power on cancer mortality. While median household income is a statistically significant predictor for both outcomes, it is a notably better predictor of cancer death ($R^2 = 0.197$) than it is of cancer diagnosis ($R^2 = 0.144$). This suggests that while wealth affects diagnosis rates, its impact on survival is even more profound.

This mathematical discrepancy suggests a "survival gap" that is dictated by wealth rather than biology. The data indicates that while Americans across various income brackets may develop cancer at somewhat comparable rates, those residing in low-income counties are nearly 20% more likely to die from the disease. This 5.3% difference in (R^2) values points directly toward systemic failures in the healthcare delivery infrastructure of impoverished regions. Specifically, the strong correlation between income and mortality identifies a lack of a financial "safety net" which manifests in several critical ways:

- **Diagnostic Delay:** In lower-income areas, the lack of affordable preventive screenings—such as mammograms or colonoscopies—often results in cancers being detected at Stage III or IV, when treatment is less effective and mortality rates are exponentially higher.
- **Treatment Barriers:** Even when diagnosed, individuals in low-income counties may lack the health insurance or liquid assets required to afford advanced oncological care, including specialized surgery, radiation, or targeted chemotherapy.
- **Socioeconomic Stressors:** Poverty is often accompanied by environmental stressors, such as proximity to industrial pollutants or "food deserts" lacking nutritional variety, which can weaken a patient's physiological resilience during aggressive cancer treatments.

V. Conclusion

This research has established a statistically significant link between socioeconomic variables and cancer outcomes across the United States through the application of simple linear regression. We demonstrated that economic indicators—particularly median household income and poverty rates—serve as potent predictors of health disparities, with a notably stronger impact on mortality than on incidence. Our findings quantify the "survival gap" revealing that while the biological risk of cancer is widespread, the probability of surviving the disease is heavily stratified by wealth and insurance coverage. Ultimately, this study underscores that addressing the cancer burden in the U.S. requires not only medical advancement but also policy-driven interventions

aimed at improving healthcare access and economic equity in underserved communities.

Future Directions

While Simple Linear Regression is a powerful start, it is limited. Future research should utilize **Multiple Linear Regression** to include variables such as:

1) Racial demographics (to identify intersections of income and race).
2) Education levels and their impact on health literacy.

By adding these variables, we can move from simple correlation to a more nuanced ”Healthcare Equity Score” for every county in the nation.

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